Distributed [*Computing*] Systems

Syllabus

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Organization, Outline, Evaluation & Recommended Readings

Organization: Meetings

- Class is scheduled Mondays from 9AM to 11:50AM in room MSTB 118 and on Wednesdays from 9:00AM to 9:50AM in room SH 128.
- We shall cover the entire course's materials in 30 hours (Quarter of 10 weeks with 3 nominal hours per week) using 4-hours per week to finish formal lectures in 7.5 weeks.
- Homework assignments will be given as we move along and a Take Home exam will be given at the end of the 30 lecture hours.
- The final 2.5 weeks of the quarter will be devoted to researching a topic of interest, related to the course, with a final reort due on the last day of instruction.

Outline

- **Introduction**
  - Historical Perspective and the quest for speed
  - The need for Concurrency ? Where is it needed? Where can it be applied?
  - Definition of Distributed [*Concurrent Computing*] Systems
  - Architectures and Classifications
  - Expectations
  - Taxonomy of Concurrent Distributed Computers
  - Example Clusters and Exa-scale Computers
- **Interconnection Networks, Communications and the OSI Model**
  - Networks define systems
  - From Clusters to Multicores
  - Point to Point v/s Switching
  - Multi-Stage Interconnection Networks
  - The OSI Model
- **Concurrent Computing**
  - Granularity of Computations
  - Complexity of Concurrent Algorithms
  - Array Computing
    - Matrix Multiplication.
    - 2D FFT.

Outline Cont"d

- **Operating System Support**
  - **A Framework for the Study of Spatial and Temporal Scheduling**
  - **Load Balancing**
    - Load Balancing Definition 1
    - Classification of Load Balancing Algorithms
    - Example Load Balancing Algorithms
    - Load Balancing Definition 2 and Rate of Change Load Balancing
  - **Fault Tolerance and Reliability**
    - Basic Concepts: Definitions and Type of Faults
    - Redundancy, Checkpointing, Transactions
    - Algorithm-based Fault Tolerance (ABFT)
- **Other applications of Distributed Systems** (Time Permitting)
  - Data Bases
  - Transaction Processing
  - The Internet !!

Evaluation

- Homework assignments: some will require coding/lab work
- A Take-Home Exam
  - Please do explain your work !!!
  - Can consult with peers, other professors, grad students, TAs, RAs, etc.
  - BUT SUBMIT YOUR OWN WORK
- A group term research project. Groups of at least 3 students. Report due on the last day of instruction (week 10).
- Final Numerical Grade:
  - 30% (AVG of HW)
  - 30% (Take Home Exam)
  - 40% (Project Grade)
- Letter grade assigned on a curve

Materials and Contact

- Materials
  - Slides will be distributed via e-mail to the class' distribution list (enrolled students only).
  - In addition to Recommended Reading, technical/research papers will be distributed via e-mail and/or Canvas and/or referenced (respecting copyright).
  - Homework assignments and Take Home Exam Distributed via e-mail and/or Canvas.
  - Group Research projects proposals submitted via e-mail to instructor. Need to obtain approval.
  - Submit HWs, Exam and Project via Canvas.
- Contact
  - Feel free to make contact via e-mail. isaac@ics.uci.edu - Will answer promptly.
  - Office hours after class or by appointment.

Recommended Reading

- Andrew S. Tanenbaum and Marteen Van Steen, Distributed Systems: Principles and Paradigms, Prentice Hall, 2nd Edition, 2007.
- Nancy A. Lynch, Distributed Algorithms, Morgan Kaufmann Publishers Inc., 1997.
- Hagit Attiya and Jennifer Welch, Distributed Computing: Fundamentals, Simulations and Advanced Topics, McGraw Hill International, 1998.
- Frank T. Leighton, Introduction to Parallel Algorithms and Architectures: Arrays, Trees, Hypercubes ,Morgan Kaufmann.
- Dimitri P. Bertsekas and John N. Tsitsiklis, Parallel and Distributed Computation: Numerical Methods, Prentice-Hall, 1989, republished in 1997 by Athena Scientific; available for download (730 scanned pages, about 55Mb).
- Kai Hwang, Advanced Computer Architecture: Parallelism, Scalability, Programmability, McGraw Hill Book Co., New York, 1993.
- Kai Hwang and Faye Briggs, Computer architecture and parallel processing, McGraw-Hill, 1984.
- Behrooz Parhami, Introduction to Parallel Processing: Algorithms and Architectures, Plenum, New York, 1999.